

Restoration Ecology of an Endangered Plant Species: Establishment of New Populations of *Cirsium pitcheri*

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Abstract

We determined the effects of shade, burial by sand, simulated herbivory, and fertilizers on the survival and growth of artificially planted population of *Cirsium pitcheri*—an endangered plant species of the sand dunes along Lake Huron. Sand burial experiments showed that greenhouse grown plants should optimally be transplanted into areas receiving 5 cm of sand deposition: burial at this depth maximized emergence, survivorship, and below-ground biomass. Under field conditions, simulated herbivory of up to 50% of the plant height produced a slight increase in biomass after one year of growth. Field observations showed that when white-tailed deer removed more than 50% of the transplant's leaf tissue, the plant died. The application of a 20:20:20 (N:P:K) water-soluble fertilizer produced a significant increase in the dry leaf biomass, total leaf area, and total dry biomass relative to control plants. We also tested for the presence or absence of a persistent seed bank. Few seeds were recovered from soil samples collected from Pinery Provincial Park and Providence Bay. However, *C. pitcheri* has the ability to form a persistent seed bank under field conditions but only at soil depths of ≥ 15 cm. *Cirsium pitcheri* seeds are able to germinate and seedlings can emerge from a burial depth of up to 6 cm. Thus, seeds planted in open, sunny areas will probably maximize emergence, growth, and survivorship of seedlings. Populations of

C. pitcheri can be restored by planting seeds at shallow depths, transplanting greenhouse-grown plants, applying water soluble fertilizers, and protecting plants from herbivores.

Key words: *Cirsium pitcheri*, restoration ecology, burial in sand, seed bank, simulated herbivory.

Introduction

The successful restoration of a rare or endangered species requires an understanding of not only its life history and population dynamics but also the complex interactions between external and internal factors within the ecological community (Miller et al. 1992). Pavlik (1994) proposed that, for the preservation of rare and endangered species, ecologists should (1) conduct a careful study of species demography (births, deaths, survival, etc.) to assess the potential of the population to maintain its present size; and (2) identify both intrinsic and extrinsic factors that are responsible for the decline of a population. Several researchers (McEachern 1992; McEachern et al. 1994; D'Ulisse & Maun 1996; Maun et al. 1996; Phillips & Maun 1996; Maun 1997) have studied critical factors affecting the populations of *Cirsium pitcheri* ([Torr. Ex. Eaton] Torrey & Gray) (Pitcher's thistle or beach thistle) since 1990.

Cirsium pitcheri is a rosette-forming monocarpic perennial which establishes from seeds, and the plants may grow as rosettes for 3 to 10 years before flowering and dying (McEachern 1992). Both stems and lower surfaces of green leaves are densely covered with long white hairs, which give the plant a silvery white appearance. The species exists as metapopulations (a set of local populations linked together through dispersal) in sand dune systems and exhibits large yearly variability in density, both within and among metapopulations, mainly because of variable rates of seedling establishment per year and variable sizes of juvenile plants (McEachern 1992). Similarly, recruitment to metapopulations is variable across sites because of (1) variance in the number of flowering adults between years (D'Ulisse & Maun 1996); (2) variable insect predation (Stanforth et al. 1997); and (3) herbivory during vegetative and reproductive phases (Phillips & Maun 1996). Pavlovic (1994) classified *C. pitcheri* as a species that persists only under natural disturbance conditions and declines under human impact. In Canada, the species is considered endangered because of a loss of habitat, habitat degradation, and continued risk to remaining populations from recreational use and development (Maun 1999).

Soil seed banks play a significant role in the dynamics of plant communities. Buried seeds provide a source of new seedlings and can be used successfully for the

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restoration of endangered species. For example, the existing seed banks may be enhanced by artificial planting of seeds. However, the most successful method for the restoration of *C. pitcheri* populations has been transplantation of greenhouse-grown, well-established plants, because transplants show high rates of establishment and require less time to reach flowering and fruiting stages (Loveless 1984; Bowles & McBride 1996). Once established, greenhouse-grown plants are able to adapt their growth patterns to the field environment. The only drawback of this technique would be the introduction of certain genotypes in the greenhouse grown plants that would not have survived environmental selection at the seedling stage in the field (Fahselt 1988; Bowles & McBride 1996). However, even the greenhouse-grown transplants face strong environmental selection and must cope with many environmental stresses during their establishment, growth, and reproduction.

In this study, our objective was to determine the effects of three critical stress factors: (1) herbivory by white-tailed deer; (2) burial by sand; and (3) nutrient limitation, on a recently established population of greenhouse grown plants of *C. pitcheri*. We also examined the probable existence of a seed bank and the effects of shade on the establishment of seedlings of *C. pitcheri*.

Methods

Seed Bank Dynamics

Answers were sought for the following two questions: (1) do the populations of *C. pitcheri* maintain a seed

bank; and (2) would the seeds remain dormant and viable in the soil for an extended period of time?

Experiment 1: Existence of a Seed Bank. Sand cores were taken from a naturally occurring population in the Wilderness Area at Pinery Provincial Park (43°15'N, 81°50'W) and the shoreline sand dunes at Providence Bay (45°39'N, 82°16'W), Manitoulin Island, Canada during 1997 and 1998 (Table 1). At Pinery Provincial Park on both July sampling dates, transect lines were established perpendicular to the shoreline and sand cores were obtained from seven sites within the dunes (Table 1). A similar procedure was also used at Providence Bay, Manitoulin Island. In 1998, at both sites, the number of samples was decreased but sample diameter was increased. Samples taken in 1998 were located around plants that had reproduced in 1997. Sand cores were passed through a 0.5-mm sieve to remove the majority of sand particles. Detritus and litter left on the sieve were carefully examined for the presence of seeds. All *C. pitcheri* seeds were placed in petri dishes containing a layer of filter paper (Anchor Paper Co., St. Paul, MN) and 10 ml of distilled water. The dishes were then set to germinate for 30 days in an incubator (25°C day, 10°C night, 14 hour photoperiod). As seeds germinated, they were recorded and taken out of the dishes. After 30 days, the seed coats of remaining seeds were removed and embryos classified as rotten or potentially viable. All potentially viable embryos were then placed in a petri dish and set to germinate for another 30 days in an incubator (25°C day, 10°C, 14 hour photoperiod). An embryo was considered viable (germinated) when the radicle had grown to 2 mm.

Table 1. Number, dimensions, location, and sites within dunes, used to determine the existence of a seed bank at Pinery Provincial Park and Providence Bay during 1997 and 1998.

Date	Number of Cores	Dimensions of Cores	Sites within Dunes
1997 Pinery Provincial Park			
July 10	350	2.5 cm diameter 35 cm depth	1. High beach 2. I dune ridge 3. Leeward side of I dune 4. Slack 5. Windward side of II dune 6. II dune ridge 7. Leeward side of II dune
July 28	350	5 cm diameter 30 cm depth	Same seven sites as on July 10
1998 Pinery Provincial Park			
July 15	4	1 meter diameter 30 cm depth	Windward side of II dune
1997 Providence Bay			
July 5	200	2.5 cm diameter 30 cm depth	Various sites from high beach to the road
1998 Providence Bay			
July 18	10	1 meter diameter 30 cm depth	Various sites from high beach to the road

Experiment 2: Seed Bank Potential and Stratification. To determine if *C. pitcheri* had the ability to maintain a seed bank, seeds (from artificially grown plants in a garden plot) were counted into batches of 50 seeds each ($n = 100$) on 15 August 1997. Each batch was mixed with 60 g of beach sand from Pinery Provincial Park and sewn into nylon bags (8×12 cm). Treatments consisted of five retrieval dates (1 April, 15 May, 30 June, 15 August, and 1 October 1998) and four burial depths (1, 5, 15, and 25 cm), with five replications per treatment. Bags were buried in a fenced site near Picnic Area # 9 at Pinery Provincial Park on 1 October 1997. For burial, sand was removed to the appropriate depth and randomly chosen bags were placed 10 cm apart from each other. Sand was then replaced to the original sand level. Square wooden frames (20-cm tall) were placed on top of the sand to prevent sand erosion and burial. Plots were checked regularly and all weeds were removed. On each of five dates (1 April, 15 May, 30 June, 15 August and 1 October 1998), five bags were retrieved from each of four burial depths and brought back to the lab at The University of Western Ontario. The bags were opened and all 50 seeds retrieved by sifting the samples through a sieve (0.5-mm openings). Seeds were carefully examined and separated into germinated and non-germinated seeds. Ungerminated seeds were placed in petri dishes and set to germinate as in Experiment 1. At the end of the experiment there were three categories of seeds: (1) germinated in the field; (2) dead (rotten); and (3) dormant (germinated after the removal of seed coat).

Experiment 3: The Effect of Shade in Seedling Establishment. The objective of this study was to determine if providing shade to young seedlings improved their establishment. On 15 September 1997, two plots (3×2.5 m) were created at the Environmental Science Western Field Station in London, Ontario, Canada (70 km from the Pinery) by digging pits in a loam soil to a depth of one meter and then filling them with beach sand from Pinery Provincial Park. The plots were divided into 12 subplots of $2.5 \text{ m} \times 25 \text{ cm}$. The subplots were raked flat. At the south end of each plot a one-meter high snow fence was erected. The snow fence was covered with black landscape fabric to provide shade. Fabric was replaced every three months to prevent holes from developing. Each subplot was then divided into 10 quadrats of $25 \times 25 \text{ cm}$ each. Within each quadrat, 10 seeds of *C. pitcheri* were sown 2 cm below the sand surface.

During summer 1998, the date of seedling emergence, the number of emerged seedlings per quadrat and their survival were recorded regularly. To eliminate edge effects, subplots 1 and 12 from each large plot were deleted from analysis. Within each subplot, quadrats 1 through 4 were used as shade quadrats, and 7 through 10 as open quadrats. To eliminate error from the varying lengths of shadow during summer, quad-

rats 5 and 6 were not used in the analysis. The amount of photosynthetically active radiation (PAR) was measured hourly (15 readings from 7:00 a.m. to 9:00 p.m.) in both shade and open areas for three days during the summer using a Li-Cor light meter Model LI-250 (LI-COR Inc., Lincoln, NE). Simultaneously, the temperature was recorded at 1 cm above the sand surface. All plants were harvested from 22–24 August 1998, by carefully collecting all leaves and digging up all root material. The sand was washed from the plants using tap water. The plants were towel dried and then the number of leaves per plant, length of the longest leaf, root length, and total leaf area per plant were recorded. Leaf area was measured using a Li-Cor portable leaf area meter (LI-COR Inc., Lincoln, NE). Plant fractions were then placed in paper bags, oven dried to a constant weight at 80°C , and then weighed.

Experiments on Transplants

For these experiments, populations of *C. pitcheri* plants (6 months old) were artificially established in September 1996 at Picnic Area # 9 within the fenced area beside the washrooms ($n = 42$) and in a blowout ($n = 226$) in the Wilderness Area of the Pinery Provincial Park.

Experiment 4: Effects of Simulated Herbivory. On 7 August 1997, 30 plants of similar size were chosen randomly from the transplanted population of *C. pitcheri*. They were randomly assigned to one of the following three herbivory treatments: control (no herbivory), 25% and 50% of plants clipped (by height). For clipping at each height, leaves were held upright, measured with a ruler from the soil surface and then clipped from the top by 25 or 50% using a razor blade. Ten replicates were used for each treatment. Survivorship and growth (number of leaves and length of longest leaf) were recorded regularly during summers 1997 and 1998. On 4 September 1998, each plant was harvested by collecting all live aboveground and belowground plant material. The following measurements were recorded: number of leaves per plant, length of the longest leaf (cm), and total leaf area per plant (cm^2). Total cumulative leaf area per plant was measured using a Li-Cor portable leaf area meter. The above- and belowground plant material were then placed in paper bags, oven dried to a constant weight for seven days at 80°C , and then weighed.

Experiment 5: Effects of Burial by Sand. On 1 November 1997, 40 plants were chosen randomly from the transplanted *C. pitcheri* population. Each plant was then randomly assigned to one of the following treatments: control (0 cm), 5, 15, and 25 cm of burial by sand. Square ($50 \times 50 \text{ cm}$) wooden frames with variable depths (depth of each frame was 3 cm more than the burial depth), were placed

around each plant and the plants were buried to the appropriate depth. Ten replicates per treatment were used. During summer 1998, emergence, survivorship, and growth (number of leaves, length of longest leaf) of each plant were recorded regularly. On 3 September 1998, all aboveground live tissue per plant was harvested and then all belowground tissue was dug up and collected. The procedure for processing the material and collecting data was identical to that of the experiment on simulated herbivory.

Experiment 6: Application of Fertilizer. For this experiment, 40 plants of *C. pitcheri* were chosen randomly from the transplanted population. The following treatments were then imposed on 25 April 1998: control (64 ml of distilled water) per plant, 64 ml of 1 g/L of water soluble 20:20:20 (N:P:K) per plant, 64 ml of 2.8 g/L of water soluble 20:20:20 (N:P:K) per plant, and 1.6 g of slow release 14:14:14 (N:P:K) fertilizer pellets (Nutricote) per plant. Water soluble fertilizer solutions were poured in a circle 3 cm from the plant. The fertilizer pellets were buried at 2 cm below the sand surface in a circle about 5 cm from the plant. During summer 1998, survivorship and the growth (number of leaves, length of longest leaf) of each plant were recorded regularly. On 3 September 1998, all aboveground live tissue was harvested and then all belowground tissue was dug up and collected. The procedure for processing the material and recording data was identical to the Experiment on simulated herbivory.

Statistical Analysis

All statistical tests were performed using MINITAB version 11.2. The percent values in all experiments were transformed to arcsine square root to normalize the data before analysis. Student's *t*-tests were used to determine differences between shade and sun treatments

in the experiment on the effect of shade. In the experiments on simulated herbivory, burial by sand, and application of fertilizer, one- or two-way analysis of variance (ANOVA) were used to determine significance. The data for the number of leaves per plant were log transformed before the use of ANOVA.

Results

Experiment 1: The Existence of a Seed Bank

Sand samples collected from Pinery Provincial Park and Providence Bay contained very few seeds. In 1997, none of the 700 samples collected from the Pinery Provincial Park in July contained seeds. Only one dormant and three dead seeds were found in 1998. Similar results were recorded from samples collected from Providence Bay, Manitoulin Island, with only two dormant and three dead seeds in 1997 and five dormant and eight dead seeds in 1998.

Experiment 2: Seed Bank Potential and Stratification

The two-way ANOVA showed a significant effect of depth of burial and date of retrieval on the numbers of germinated, dormant and dead seeds (Table 2). There was also a significant interaction between burial depth and the date of retrieval for germinated and dormant seeds but not for dead seeds (Table 2).

The percent germination of seeds was significantly higher from 1-cm burial compared to 5-, 15-, and 25-cm burial depths on all dates of retrieval (Table 2; Fig. 1). By 1 October it was evident that seed germination decreased significantly with an increase in burial depth of seeds. Lowest germination occurred at 25-cm depth (Table 2; Fig. 1). The seedlings started to germinate and emerge from sand in late March. As the season pro-

Table 2. Two-way ANOVA for the percentages of germinated, dormant, and dead seeds of *C. pitcheri* at five dates of retrieval, four depths of burial, and interaction between burial depth and date of retrieval.

	Source of Variation	df	SS	MS	F	p
Germinated seeds	Depth of burial	3	7.104	2.368	526.56	< 0.001
	Date of retrieval	4	0.864	0.216	48.05	< 0.001
	Depth × date	12	0.126	0.011	2.33	0.013
	Error	80	0.359	0.005		
	Total	99	0.454			
Dormant seeds	Depth of burial	3	4.177	1.392	158.74	< 0.001
	Date of retrieval	4	1.470	0.367	41.86	< 0.001
	Depth × date	12	0.199	0.017	1.89	0.048
	Error	80	0.701	0.008		
	Total	99	6.545			
Dead seeds	Depth of burial	3	1.036	0.367	164.88	< 0.001
	Date of retrieval	4	0.061	0.015	6.84	< 0.001
	Depth × date	12	0.004	0.000	0.16	0.999
	Error	80	0.178	0.002		

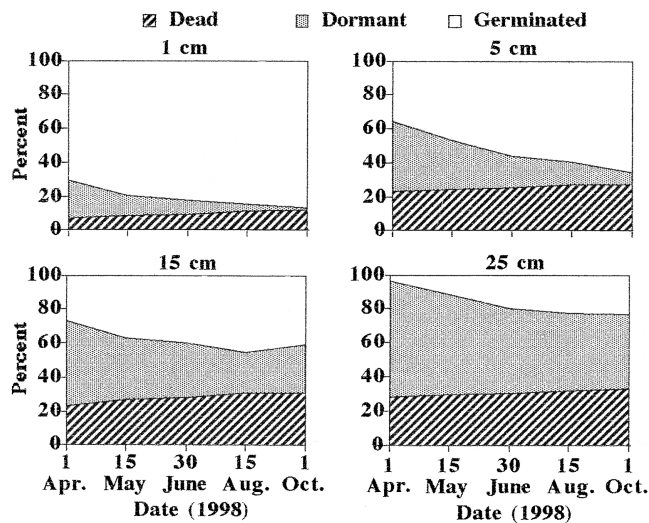


Figure 1. Percentage of *C. pitcheri* seeds that had germinated, were dormant and viable, or had died at various depths of burial in sand (1, 5, 15, and 25 cm). The seeds were retrieved on five dates after burial.

gressed the number of germinated seedlings increased in all burial depth treatments (Fig. 1).

The largest percentage of dormant seeds was at the 25-cm burial depth on 1 April, and the lowest at 1-cm depth on 1 October (Fig. 1). The percent of dormant seeds increased significantly with an increase in depth of burial on each date of retrieval (Table 2). However, within each burial depth the percent of dormant seeds significantly decreased over time (Fig. 1).

There were a significantly lower number of dead seeds at 1 cm as compared to other burial depths for all dates of retrieval (Fig. 1; Table 2). The percentage of dead seeds increased significantly as the season progressed with later dates of retrieval at the 1-cm burial depth (Fig. 1). For all other burial depths the percentage of dead seeds did not differ significantly among dates of retrieval.

Experiment 3: The Effect of Shade on Seedling Establishment

The shaded area received a total PAR $\pm$ SE (sum total of 15 readings taken each hour from 7:00 a.m. to 9:00 p.m.) of $1,254 \pm 37.6 \mu\text{mol m}^{-2} \text{s}^{-1}$ per day ($n = 3$) compared to $9,915.3 \pm 166.8 \mu\text{mol m}^{-2} \text{s}^{-1}$ in the open area. The air temperature (averaged over three days) in the shaded area at solar noon (2:00 p.m.) was also lower ($28.7 \pm 0.67^\circ\text{C}$) than in the open ($42.3 \pm 1.9^\circ\text{C}$).

There were significantly higher numbers of emerged seedlings per quadrat in the open (mean = 4.8 ± 0.3) compared with the shaded area (mean = 3.5 ± 0.2) ($t = -3.3$, $df = 152$, $p < 0.001$). Similarly, the number of surviving seedlings per quadrat was significantly higher in the open

(mean = 4.1 ± 0.3) compared to the shaded areas (mean = 3.0 ± 0.2) ($t = -2.94$, $df = 157$, $p < 0.003$). Survivorship curves showed that fewer seedlings survived in the shaded treatment compared with the open area of the plot (Fig. 2). Seedlings growing in full sunlight had significantly larger total leaf area per plant, root length, length of the longest leaf, number of leaves per plant, and total dry biomass of leaves, roots, and the whole plant than those growing in the shade according to Student's t -test (Table 3).

Experiment 4: Effects of Simulated Herbivory

All 30 plants subjected to herbivory treatments survived until they were harvested in September 1998. The mean number of leaves did not differ significantly between the control and the burial treatments except on 26 August when there was a significant decrease in the number of leaves per plant in the 50% herbivory treatment (data not presented). However, the mean length of the longest leaf per plant showed a significant decrease over control within one week of both the 25 and 50% herbivory treatments. The plants continued to show significant differences among all treatments for the remainder of the 1997 summer following treatment application. By April 1998, all plants subjected to both herbivory treatments had recovered completely (Fig. 3). There were no significant differences between treatment means in the length of longest leaf on any date during the 1998 summer. The herbivory treatments did

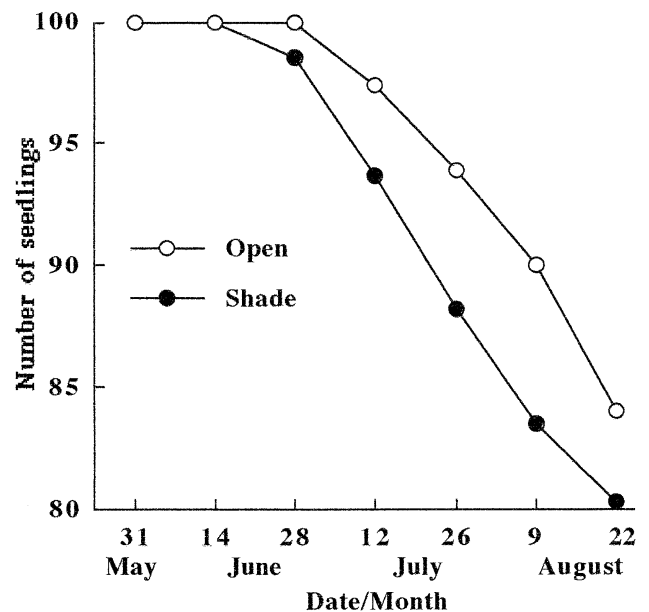


Figure 2. Survivorship of *C. pitcheri* seedlings over time in shaded and open areas on different dates during summer 1998.

Table 3. Comparison of growth parameters of seedlings of *C. pitcheri* growing in the shade versus the open, along with the mean values, standard errors, and the probabilities according to Student's *t*-test.

Plant Parameter	Habitat	Mean	Standard Error	"t" value	Probabilities
Total leaf area per plant (cm ²)	Shade	40.58	2.6	-6.3	<0.0001
	Open	65.89	3.1		
Root length (cm)	Shade	21.2	0.6	-7.0	<0.0001
	Open	28.3	0.8		
Length of longest leaf (cm)	Shade	16.2	0.7	-2.7	<0.009
	Open	18.9	0.8		
Number of leaves per plant	Shade	5.4	0.1	-4.6	<0.0001
	Open	6.3	0.1		
Total dry leaf biomass per plant (g)	Shade	0.37	0.032	-2.4	<0.02
	Open	0.49	0.038		
Total dry root biomass per plant (g)	Shade	0.26	0.031	-3.3	<0.001
	Open	0.42	0.035		
Total dry biomass per plant (g)	Shade	0.64	0.044	-4.1	<0.0001
	Open	0.91	0.050		

not significantly decrease the total leaf area per plant 13 months after the defoliation treatments (Fig. 3). Similarly, there were no significant differences in dry biomass of roots, leaves, and total biomass per plant between control and the two herbivory treatments (Fig. 3). If anything, there was a slight stimulation (not signifi-

cant) in plant growth following herbivory of plants during the 1998 growing season.

Experiment 5: Effects of Burial by Sand

All plants emerged and survived burial depths of 0, 5, and 15 cm. However, only five of ten plants emerged from the 25-cm burial depth and of these, only three survived until September 1998. The date of emergence in spring was related to burial depth. The greater the burial depth, the slower was the rate of emergence of plants. All plants in the 0, 5, and 15 cm burial treatments emerged by 10 May 1998, except two plants buried at 15-cm depth whose emergence was delayed by four weeks. In contrast, only three of the ten plants had emerged from the 25 cm burial depth by 10 June and two more grew out of the sand between 10 June and 25 June.

The number of leaves per plant was significantly higher in the control and 5-cm burial depths than 15- and 25-cm burial treatments until about 10 June (Table 4). Thereafter, the 15- and 25-cm treatments started to catch up. By 25 July, the number of leaves in the 15- and 25-cm treatments had caught up to the control and there were no significant differences among treatment means. When the plants were harvested on 3 September, the plants in both the 15- and 25-cm treatments had significantly greater numbers of leaves per plant than the control plants (Table 4).

On 25 April, the leaves were significantly longer in the 0- and 5-cm burial treatments than in the 15- and 25-cm burial treatments (Table 4). The length of longest leaf was significantly lower in the 25-cm burial treatment than in the other treatments until the end of July. By the end of the summer (3 September), there was no longer a significant difference between the 15- and 25-cm treat-

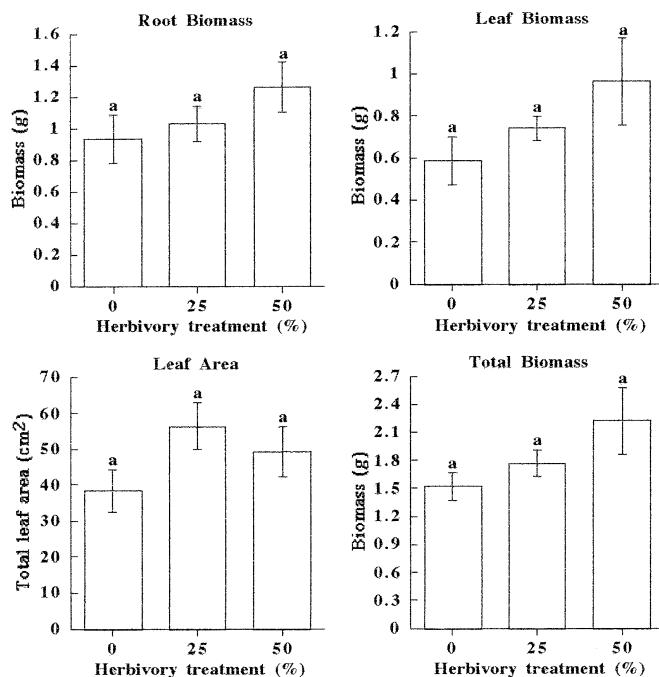


Figure 3. The effects of simulated herbivory treatments (0% [control], 25%, and 50% clipping of plants by height) on total leaf area per plant, and leaf, root, and total biomass per plant of *C. pitcheri*. Vertical lines represent one standard error. Means within the same graph with different superscript letters just above the standard error bars are significantly different ($p < 0.05$).

Table 4. The mean (SE) number of leaves and the length of the longest leaf (cm) during the summer 1998 for plants buried to 0 cm (control), 5, 15, and 25 cm depths in sand during winter 1997. Treatment means within the same date, followed by a different letter, are significantly different at $p < 0.05$.

Date	Burial Depth (cm)	Number of Leaves per Plant	Length of Longest Leaf (cm) per Plant
25 April	0	4.5 (0.40) ^a	8.1 (0.77) ^a
	5	5.1 (0.23) ^a	8.2 (0.55) ^a
	15	1.6 (0.60) ^b	4.1 (1.42) ^b
	25	0.2 (0.20) ^b	0.3 (0.30) ^c
10 May	0	5.2 (0.33) ^a	13.5 (1.09) ^a
	5	6.5 (0.27) ^a	14.2 (0.51) ^a
	15	2.6 (0.64) ^b	10.3 (1.89) ^a
	25	1.0 (1.00) ^b	0.7 (0.70) ^b
25 May	0	5.7 (0.30) ^a	16.6 (1.21) ^a
	5	7.2 (0.25) ^a	18.9 (0.73) ^a
	15	3.6 (0.43) ^b	13.9 (2.10) ^a
	25	2.6 (1.25) ^b	2.1 (1.40) ^b
10 June	0	6.0 (0.30) ^{ab}	17.3 (1.16) ^a
	5	7.9 (0.31) ^a	18.7 (1.04) ^a
	15	4.9 (0.50) ^b	17.0 (1.44) ^a
	25	4.6 (1.63) ^b	3.9 (2.29) ^b
25 June	0	6.6 (0.48) ^b	16.9 (1.53) ^a
	5	8.6 (0.22) ^a	20.4 (1.14) ^a
	15	5.8 (0.49) ^b	18.9 (1.17) ^a
	25	5.2 (1.28) ^b	5.6 (2.29) ^b
10 July	0	7.1 (0.53) ^{ab}	16.8 (1.17) ^a
	5	9.0 (0.15) ^a	20.5 (1.39) ^a
	15	6.5 (0.52) ^b	16.7 (1.34) ^a
	25	6.4 (1.29) ^b	7.0 (3.52) ^b
25 July	0	7.5 (0.62) ^a	15.7 (1.67) ^a
	5	9.3 (0.26) ^a	20.5 (1.48) ^a
	15	7.4 (0.69) ^a	15.8 (1.90) ^a
	25	7.2 (1.83) ^a	9.0 (4.59) ^b
3 Sept	0	5.5 (0.48) ^b	16.3 (1.33) ^{ab}
	5	6.3 (0.68) ^{ab}	19.9 (1.66) ^a
	15	7.8 (0.76) ^a	11.8 (1.25) ^b
	25	11.7 (2.40) ^a	9.2 (0.67) ^b

ments and the control. The only significant difference in the mean length of the longest leaf was between the 5-cm and both the 15- and 25-cm burial treatments (Table 4). The belowground biomass per plant was significantly lower in the 15- and 25-cm burial treatments compared to the control and 5-cm burial. When the plants were buried at 15- and 25-cm depths, the apical meristem grew through the sand and produced a new rosette at the new soil surface. Thus, the length of the belowground stem and root was significantly greater in plants buried to 15- and 25-cm depths than control and 5-cm burial treatments (Fig. 4). There was no difference in the leaf biomass per plant between the control and the 5- or 15-cm burial depths; however, plants buried at 25 cm had significantly larger biomass than control plants. The total leaf area per plant was significantly greater in all burial treatments than in the control plants. As the depth of burial increased, so did the leaf area per plant. By the end of the experiment, the total leaf area in the 25-cm burial treatments was almost four times that of

the control plants. The total biomass per plant did not differ among all burial treatments.

Experiment 6: Application of Fertilizer

The number of leaves per plant was unaffected by fertilizer treatments during the summer of 1998. The length of the longest leaf per plant was not significantly different between treatment means on 25 April, immediately following treatment application. However, by 10 May there was a significantly lower value for longest leaf length in plants receiving the slow release fertilizer pellets in comparison to control. This trend continued until the end of summer (3 September 1998). The other two fertilizer treatments significantly increased leaf length over both the control and slow release fertilizer pellet treatments by 10 June 1998. Fertilizer concentrations of 1 g/L and 2.8 g/L were both very effective in increasing leaf length, but the 2.8 g/L treatment was significantly

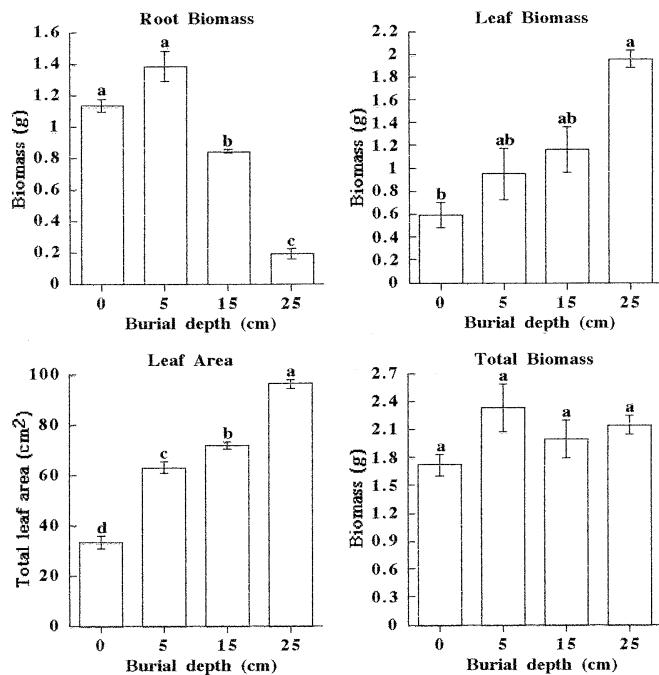


Figure 4. The effects of burial treatments (0, 5, 15, and 25 cm burial depths) on total leaf area per plant, and leaf, root, and total biomass per plant of *C. pitcheri*. Vertical lines represent one standard error. Means within the same graph with different superscript letters just above the standard error bars are significantly different ($p < 0.05$).

better than the 1.0 g/L and all other treatments after 10 July.

When the plants were harvested, a significantly lower total dry root biomass per plant was recorded for plants receiving 1.6 g of fertilizer pellets, compared to the other treatments which did not differ significantly from one another (Fig. 5). Plants receiving 1.0 g/L of water-soluble fertilizer had longer roots than those in the control and 1.6 g fertilizer pellet treatments. The mean total dry leaf biomass per plant in the 2.8 g/L of water-soluble fertilizer treatment was greater than in the control and 1.6 g fertilizer pellet treatments (Fig. 5). The mean total leaf area per plant was significantly higher in both water-soluble fertilizer treatments than in the control and 1.6 g fertilizer pellet treatments (Fig. 5). There was a significantly lower mean total leaf area per plant for the slow release fertilizer treatment (1.6 g fertilizer pellets) compared to the control. The results for total dry biomass per plant were similar to those for dry leaf biomass per plant. There was a significantly higher total dry biomass in the 2.8 g/L of water-soluble fertilizer treatments compared to the control. Plants given the slow release fertilizer pellet treatment had the least total dry biomass (Fig. 5).

Discussion

Seed Bank

A seed bank, defined as viable seeds present both above and below the soil surface (Thompson & Grime 1979), is potentially important to the dynamics of an endangered plant species because it can maintain the population over time (Pavlik et al. 1993; Teo-Sherrell et al. 1996), and can serve as a pool of genetic material that enables a species to withstand a range of responses to ordinary environmental conditions (Teo-Sherrell et al. 1996). Does *C. pitcheri* maintain a persistent seed bank [seeds that are viable for more than one year (Thompson & Grime 1979)] in its natural field populations? There are strong indications from this study that a persistent seed bank may not be a contributing factor in the population growth of *C. pitcheri*. These results are consistent with those found at Sleeping Bear Dunes National Lakeshore, along Lake Michigan, in an area containing a large natural population of *C. pitcheri* (S. Louda 1998, University of Nebraska, Lincoln, personal communication). A similar conclusion was also reached by Baptista and Shumway (1998) who showed that the existing seed banks in four different coastal sand dunes were very small and hence unlikely to enhance species diversity during restoration.

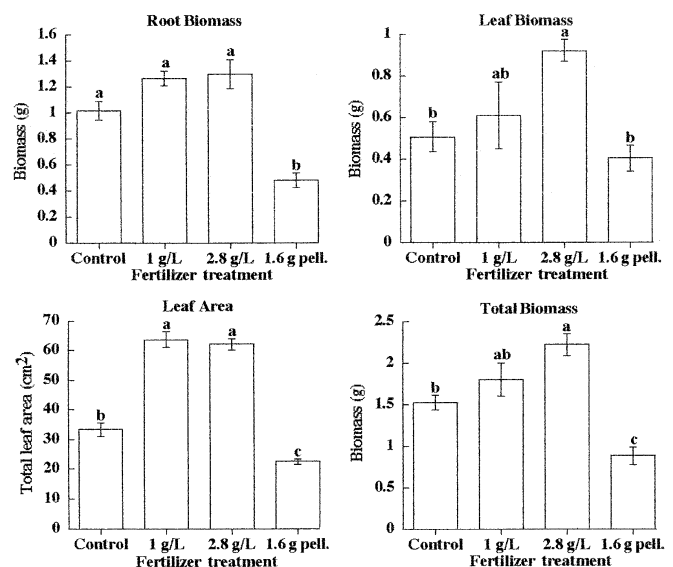


Figure 5. The effects of fertilizer treatments (control [distilled water], 1 g/L and 2.8 g/L of water soluble 20:20:20 [N:P:K] per plant and 1.6 g of slow release 14:14:14 [N:P:K] fertilizer pellets) on total leaf area per plant, and leaf, root, and total biomass per plant of *C. pitcheri*. Vertical lines represent one standard error. Means within the same graph with different superscript letters just above the standard error bars are significantly different ($p < 0.05$).

Does *C. pitcheri* have the potential to maintain a persistent seed bank? The data indicated that *C. pitcheri* does have the potential to have a small persistent seed bank because after one year, more than 1, 6, 27, and 43 percent of the seeds had not germinated but remained viable at 1-, 5-, 15-, and 25-cm depths, respectively. The optimal depth at which seeds formed a persistent seed bank was more than or equal to 15 cm since a minimum of 28% of the seeds remained dormant at 15 and 25 cm until October. The greater the burial depth, the greater was the proportion of seeds that remained dormant. Similar results were reported by Lee and Ignaciuk (1985) for seed germination of three species (*Atriplex laciniata* Torr., *Cakile maritima* Scop. and *Salsola kali* L.); by Doucet and Cavers (1996) for seed germination of *C. vulgare*, and by Zhang and Maun (1994) for seed germination of seven species on lacustrine sand dunes of Lakes Huron and Erie. Thus, the microenvironment of deeply buried seeds is not conducive to their germination and prevents them from premature germination.

Would these seeds contribute to the future growth of the population and aid in the long-term survival of the species? Since they are deeply buried, erosion of sand would have to occur before the seedlings from germinated seeds would be able to emerge above the sand surface. Chen and Maun (1999) showed that the maximum depth from which seedlings of *C. pitcheri* could emerge was 6 cm. Hence, the erosion of at least 9 to 19 cm of sand would have to take place from 15- and 25-cm depths, respectively, before dormant seeds would contribute to the population. Thus, the contribution of a persistent seed bank to the population is questionable. Although the species has the potential to maintain a seed bank, the seeds must be buried deeply at greater than 15- to 25-cm depths. A transient seed bank (seeds are viable for less than one year) could be formed each year at depths of less than 6 cm (Chen & Maun 1999). This would aid in the growth of the populations, while allowing natural selection to operate on seedlings.

The Effect of Shade in Seedling Establishment

Nurse-plants are known to improve survival of seedlings in their vicinity (McLeod & Murphy 1977; Kellman & Kading 1992). Shade cast by nurse-plants lowers surface soil temperatures and reduces evapo-transpiration of young seedlings, which allows them to escape desiccation. Does *C. pitcheri* require nurse-plants for increased survival and establishment of seedlings in sand dunes? Whereas the presence of nurse-plants might be useful for increased seedling survival of some plant species such as *Ptelea trifoliata* (McLeod & Murphy 1977), *Calamovilfa longifolia*, or *Artemisia caudata* (Maun 1990), it is doubtful that it would benefit *C. pitcheri* because seedlings grown in the shade exhibited lower sur-

vival, produced fewer leaves per plant and showed lower biomass per plant. Clearly, *C. pitcheri* is a heliophyte that requires full sunlight for maximum survival and growth. Thus, for artificial increase of its populations the species should be planted in areas receiving full sunlight.

Herbivory by White-Tailed Deer

Transplanted populations of *C. pitcheri* must survive many natural causes of mortality. Within Pinery Provincial Park, one of the leading causes of *C. pitcheri* mortality has been herbivory by white-tailed deer (Phillips & Maun 1996). The data showed that *C. pitcheri* plants exhibited a temporary set-back soon after the imposition of herbivory but the plants had sufficient time to recover and compensate for lost tissue by the end of the next year of growth. Similar results were obtained by Phillips and Maun (1996), who showed that juvenile plants (90- and 125-day-old) compensated for limited amounts of tissue removal. However, regardless of the amount of tissue removed, the re-growth of shoots occurred at the expense of belowground biomass (Phillips & Maun 1996). Since the probability of flowering is dependent on the amount of stored root reserves (Keddy & Keddy 1984), plants subjected to bouts of herbivory would require a longer juvenile period to reach the reproductive stage, thus increasing their period of vulnerability to herbivory, insect attack, excessive burial, and other environmental stresses.

Burial by Sand

Another important factor controlling the population dynamics of *C. pitcheri* populations is recurring sand burial resulting from wind, wave action, and recreational activities (Maun et al. 1996). Burial in sand had both negative and positive effects on *C. pitcheri*. In small amounts, it stimulated plant growth and in large amounts it inhibited both emergence and survival. The increase in aboveground tissue came at the expense of belowground biomass. Following the burial treatments, experimental plants were able to increase their growth rates so that they were either equal to or surpassed control. For emergence through a sand deposit, the plant requires energy reserves in which the amount available is dependent on the season and the stage at which a plant is buried (Maun et al. 1996). Growth stimulation as a result of small amounts of burial in sand may be a result of improved soil conditions (Disraeli 1984), mycorrhizal fungi association (Little & Maun 1996), and lowered water stress (Olson 1958). When plants are buried in sand there is a decrease in both the temperature and concentration of oxygen in the root zone, but there is an increase in the amount of moisture (Maun

1998). These factors may have allowed an increase in the growth of plants.

Mineral Nutrition

The dune systems in general are deficient in three macronutrients: nitrogen, phosphorus, and potassium (Baldwin & Maun 1983). The addition of water-soluble fertilizer had a positive effect on *C. pitcheri*, showing that the application of additional nutrients improved the growth of plants. Since nutrients are quickly leached from the sand dunes, water-soluble fertilizers should be applied early in the growing season to improve growth. In general, the application of fertilizers is both expensive and time-consuming, and therefore should only be used as a short-term conservation method. Once the plant population is large enough it must be able to survive on its own. In contrast, slow release fertilizer pellets were found to be detrimental to both the growth and biomass of *C. pitcheri*, probably due to very low levels of precipitation during summer 1998.

Based on these studies, we propose that in the short term the population size of *C. pitcheri* along Lake Huron can be increased by (1) planting seeds and creating a transient seed bank in open sunny area of the sand dunes; (2) using greenhouse-grown transplants to create new populations; (3) applying water-soluble fertilizers early in the growing season; and (4) protecting plants from herbivores such as the white-tailed deer. It must be emphasized that sand dune habitats for establishing transplanted populations be carefully selected. The best approach would be to choose areas that approximately mimic the physical and biological conditions prevailing in the naturally-occurring populations.

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